

Week 5 / HW 5

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prob 7 / ex 7

$$f(x, y, z) = 2x + 2y + z \quad x^2 + y^2 + z^2 = 9$$

$$g(x, y, z) = x^2 + y^2 + z^2 = 9$$

$$\nabla f = \lambda \nabla g \rightarrow (2, 2, 1) = (2\lambda x, 2\lambda y, 2\lambda z)$$

$$2\lambda x = 2, \quad 2\lambda y = 2, \quad 2\lambda z = 1$$

$$x = \frac{1}{\lambda}, \quad y = \frac{1}{\lambda}, \quad z = \frac{1}{2\lambda}$$

possible extreme values at

$$\text{points: } \begin{matrix} \nearrow (2, 2, 1) \\ \searrow (2, -2, -1) \end{matrix}$$

$$\text{max value} \Rightarrow f(2, 2, 1) = 8$$

$$\text{min value} \Rightarrow f(-2, -2, -1) = -8$$

prob 13 / ex 18

$$\iint_D (x^2 + 2y) dA \quad D = \begin{cases} y=x \\ y=x^3 \\ x \geq 0 \end{cases}$$

$$\begin{aligned} \iint_D (x^2 + 2y) dA &= \int_0^1 \int_{x^3}^x (x^2 + 2y) dy dx \\ &= \int_0^1 \left[x^2 y + y^2 \right]_{y=x^3}^{y=x} dx = \int_0^1 (x^3 + x^2 - x^5 - x^6) dx = \end{aligned}$$

$$= \left[\frac{1}{4} x^4 + \frac{1}{3} x^3 - \frac{1}{6} x^6 - \frac{1}{7} x^7 \right]_0^1 = \frac{23}{84}$$

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$$\int_0^1 \int_x^1 e^{\frac{x}{y}} dy dx = \int_0^1 \int_0^y e^{\frac{x}{y}} dx dy = \int_0^1 \left(y \cdot e^{\frac{x}{y}} \right)_{x=0}^{x=y} dy$$

2)

$$dy = \int_0^1 (e-1) y dy = \left(\frac{1}{2} (e-1) y^2 \right) \Big|_0^1 = \frac{1}{2} (e-1)$$

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dx =